

VIL SUBSYSTEM REQUIREMENTS SPECIFICATION

Virtual Interlocking
Digital Interlocking OBB

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LOG OF CHANGES			
Revision	Date	Author	Change Description
01 RL	2025-12-12	J. Mach	First edition. Details see 5.1

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1. INTRODUCTION

1.1 PURPOSE

The present SSS document contains the specification of subsystem Virtual Interlocking (VIL), which is developed in the context of the Generic Application Digital Interlocking OBB (DIL).

Please note the following limitations of this version:

- The current requirements are preliminary (status “draft” or “in negotiation”).
- The traceability to the system level requirements is not yet stable.

Please note: If a DIL system requirement specified in [DIL_SSS_CU1] is allocated to subsystem VIL without any refinement, it will not be repeated in this VIL-SSS but still be applicable for subsystem VIL. Therefore, the complete VIL specification consists of VIL-SSS requirements as well as the DIL-SSS requirements allocated to VIL.

1.2 INTENDED AUDIENCE

- Product Design Authority Solution Element
- Solution Engineering Manager Solution Element
- Technical Project Manager Generic Application
- Solution Engineering Manager Generic Application
- Product Line Manager
- Test Manager
- Validator
- Verifier
- Safety Assurance Manager
- Quality Assurance Manager
- RAM Assurance Manager
- System Architect
- Work Package Manager
- Project Security Manager

1.3 DISTRIBUTION

A new version of the document is distributed to all persons according to chapter 1.2.

1.4 DOCUMENT MANAGEMENT

See Requirements Management Plan [DIL_RMP] for a description of the SSS document management.

1.5 STRUCTURE OF THE REQUIREMENTS

See [DIL_RMP] for a description of the foreseen SSS requirement structure.

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1.5.1 TARGET RELEASES

Requirement attribute “Target Release” can have the following values:

- VIL OBB 1.0.0: Variant “Basic Virtualization” - Phase 1
- VIL OBB 1.1.0: Variant “Basic Virtualization” - Phase 2
- VIL OBB 1.2.0: Variant “Basic Virtualization” - Phase 3
- VIL OBB 2.0.0: Variant “Enhancement Geo-Redundancy”
- VIL OBB 3.0.0: Variant “Enhancement Independence from Server Hardware”

1.6 INFORMATION FOR DOCUMENT GENERATION

The present document was generated by using IBM™ Rational DOORS.

Settings applied:

DOORS/RMF Version:	DOORS 9.7.2.6/RMF 6.1.0.9
DOORS/RMF Module Name:	EL4 SSS
DOORS/RMF Module Version (Baseline):	1.3
DOORS/RMF Module Path:	/ELEKTRA 4.0/Specification
DOORS/RMF Export View:	WEXP SSS 6955 (VIL)
DOORS/RMF Current Filter:	(TA Req Status != Obsolete) AND (TA SRS includes 6955)
DOORS/RMF Export Options 1:	2,false,0,0,false,false,0,true,false,
DOORS/RMF Export Options 2:	true,false,true,2,false,false,true,PostprocessAllStyles,\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_0011.log,
DOORS/RMF Export Options 3:	\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_0011.log,true,true,false,false,false,false,0,true,4,
DOORS/RMF Export Options 4:	3,###,===,true,false,false,false,false,
Documentation-Vob Template Name:	3BU_16000_6955_DTAPA-Template

Table 1: Document generation settings

1.7 OPEN POINTS

Please note the following limitations of this version:

- The overall structure of solution element VIL may change, to accommodate the Safety Plan.
- The input DIL system specifications have not yet been released completely. Changes have to be expected.
- The input DIL system architecture has not yet been released. Several architecture topics are still open.
- In particular the requirement attributes Origin, Verification and Allocation are preliminary and might be changed in later versions.

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2. REFERENCES AND TERMS

2.1 ELEKTRA DOCUMENTS

ID	Title	Reference / Date	Version
[EL2_GLOSS]	Glossary: Definitions, Acronyms and Abbreviations - ELEKTRA 2.x System	3BU 15000 0081 DCAPA	06
[HIT_GLOSS]	Hitachi Glossary (Latest update: http://trss-at.at.gts/glossword)	3BU 15000 8008 WDAPQ	04
[VIL_SEMP]	Subsystem Engineering Management Plan (SEMP) – Virtual Interlocking OBB Phase 1	3BU 15000 6955 DPAPA	01
[DATLOGIF]	Datalogger Interface Design Description	3BU 15000 6852 PBAPA	01 (planned)
[DIL_RMP]	Requirements Management Plan – Digital Interlocking OBB	3BU 17300 1207 DUAPA	01
[DIL_SSS_GEN]	System Requirements Specification – Digital Interlocking (DIL) – General Requirements	3BU 17300 1251 DTAPA	01
[DIL_SSS_CU1]	System Requirements Specification – Digital Interlocking (DIL) - Central Unit Part 1	3BU 17300 1255 DTAPA	01
[DIL_SSS_FE]	System Requirements Specification – Digital Interlocking (DIL) - Field Element Controller and Stations	3BU 17300 1255 DTAPA	01
[DIL_SSS_RAMs]	System Requirements Specification – Digital Interlocking (DIL) - RAMS	3BU 17300 1257 DTAPA	01
[DIL_SSDD]	System Design Specification – Digital Interlocking (DIL)	3BU 15000 1254 DSAPA	01 (planned)
[DIL_TOPLVL]	Top Level Architecture Drawing – Digital Interlocking (DIL)	3BU 17300 1218 DSAPA	02
[VIL_SSS_SEC]	Security Requirements Specification – Virtual Interlocking (VIL)	3BU 15000 6980 DTAPA	01 (planned)

Table 2: Referenced ELEKTRA documents

2.2 ÖBB SPECIFICATIONS

ID	Title	Reference / Date	Version
[LH-DSTW_D2.5]	Lastenheft DSTW - D2.5 Teillastenheft Teilsystem DSTW-ZE	18.12.2024	V1.3
[PH_X25]	Pflichtenheft X25 - Schnittstelle ÖBB	10.10.2022	2.1
[PH_X25-BFZ]	Pflichtenheft X25 - BFZ Schnittstelle ÖBB	03.11.2009	2.0
[LH_EBO2-X25]	Lastenheft EBO2 / X25	01.04.2022	1.0
[LH_ETCS_D1]	Lastenheft ETCS - Streckenausrüstung D.1	30.06.2021	1V3
[X25-over-IP]	Schnittstellenbeschreibung X25-over-IP/RaSTA	26.06.2020	2.0

Table 3: Referenced ÖBB specifications

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2.3 EULYNX SPECIFICATIONS

ID	Title	Reference / Date	Version
[Eu.Doc.9]	EULYNX Glossary	19.06.2025	2.9 (1.A)
[Eu.Doc.33]	Interface Specification SCI-LS	19.06.2025	4.3 (1.A)
[Eu.Doc.37]	Signal Aspect Table	19.06.2025	2.2 (2.H)
[Eu.Doc.38]	Interface Specification SCI-P	19.06.2025	4.2 (2.A)
[Eu.Doc.41]	Requirements Specification SCI-ILS	19.06.2025	4.3 (1.A)
[Eu.Doc.42]	Interface Specification SCI-ILS	19.06.2025	4.4 (1.A)
[Eu.Doc.44]	Interface Specification SCI-TDS	19.06.2025	4.1 (1.A)
[Eu.Doc.46]	Interface Specification SCI-IO	19.06.2025	4.0 (4.A)
[Eu.Doc.47]	Requirement Specification SCI-RBC	19.06.2025	4.0 (4.A)
[Eu.Doc.48]	Interface Specification SCI-RBC	19.06.2025	4.0 (2.A)
[Eu.Doc.49]	Requirements specification SCI-CC	19.06.2025	4.2 (2.A)
[Eu.Doc.50]	Interface specification SCI-CC,	19.06.2025	4.3 (2.A)
[Eu.Doc.93]	Interface specification SCI Generic,	19.06.2025	3.3 (1.A)
[Eu.Doc.100]	Specification of Point of Service - Signalling	19.06.2025	2.2 (2.A)
[Eu.Doc.111]	Requirements Specification SCI-LX	19.06.2025	2.2 (2.A)
[Eu.Doc.112]	Interface Specification SCI-LX	19.06.2025	2.1 (1.A)

Table 4: Referenced EULYNX specifications (Baseline 4 Release 4)

2.4 STANDARDS AND PROCEDURES

ID	Title	Reference / Date	Version
[EN-50716]	European Standard "Railway applications – Requirements for software development"		EN 50716:2023
[EN-50129]	European Standard "Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signaling"		EN 50129:2018 + AC:2019
[RASTA]	Sicheres Übertragungsprotokoll RaSTA gemäß EN50159 (VDE V 0831-200)		2015-06

Table 5: Standards and Procedures

2.5 TERMS, ABBREVIATIONS AND ACRONYMS

See Definitions, Acronyms and Abbreviations [EL2_GLOSS] and [HIT_GLOSS]. Please be also aware of the EULYNX Glossary [Eu.Doc.9]. Additional abbreviations: See table below.

CDS	Cluster Deployment Supervisor (component of TAS Platform)
CE	Computing Element (TAS Platform)
CG	Computing Group (TAS Platform)
CN	Computing Node (TAS Platform)
DIL	Digital Interlocking (System Level)
DIL-CU	Digital Interlocking Central Unit (German: DSTW-ZE)
JSON	JavaScript Object Notation file format
KVM	Kernel-based Virtual Machine (virtualization module in Linux kernel)

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NMR	N times M-out-of-M Redundancy (software component of TAS Platform Core)
PTP	Precision Time Protocol
SIL	Safety Integrity Level
SRAC	Safety-related Application Condition
SSDD	System/Segment Design Document
SSS	System/Segment Specification
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol
VIL	Virtual Interlocking (Subsystem Level)

Table 6: Abbreviations and Acronyms

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3. OVERVIEW

3.1 BASIC TERMS

The following terms are fundamental for the understanding of many VIL requirements. The proposed definitions will be moved to the DIL glossary after review.

3.1.1 VIL REDUNDANT NODE

A “VIL redundant node” is a set of SW components that perform the VIL functions and provide safety. It forms the basic element of a redundant VIL. A VIL redundant node is deployed on several virtual machines.

Note: A VIL redundant node is not synonymous with a TAS Platform Computing Node (CN) as foreseen for component ELEKTRA Core, as VIL also includes component EC-MUX that will use a different redundancy implementation.

3.1.2 VIL INSTANCE

A “VIL instance” is a group of synchronized VIL redundant nodes that run synchronously in order to provide high availability based on hot-hot redundancy. Each VIL instance is configured by station data to control a certain railway station. The DIL system will typically comprise many VIL instances.

3.2 CONTEXT

An overview of the DIL system including the role of subsystem VIL can be found in [DIL_SSDD].

3.3 ARCHITECTURE OVERVIEW

An overview of the VIL architecture including its main functions and interfaces as well as components will be added here.

A preliminary top-level architecture diagram showing both the DIL system as well as the VIL main components and interfaces can be found in [DIL_TOPLVL].

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4. REQUIREMENTS

4.1 VIRTUALIZATION

EL4-VIL-SSS-REQ-00215		VIL Virtualization		(Draft)	
The VIL shall run on Virtual Machines (VMs) in the cloud environment of the DIL-CU data center.					
Comments:					
Category:	FUR	Priority:	Mandatory	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00437] Architecture - Number of VIL Instances per Server ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00448] DIL Variants - COTS Hardware	
Originator:	Technics (R&D)	Origin:			
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System		
Allocation:	VIL ELEKTRA Core ECMUX	Verification:	Analysis Integration test		
		See also:			
Issues/Decisions:					

EL4-VIL-SSS-REQ-01445		VIL Instance Delivery		(Draft)
Every VIL instance shall be delivered as a set of versioned filesystem images as well as a build tool that allows combining these filesystem images into the final VM images of VIL. There shall be separate filesystem images for at least the following data:				
• TAS Platform Core • VIL application software • VIL station specific configuration data				
The delivery shall include				
• a complete list of filesystem image versions and their checksums • the checksums of the final VM images				
The build tool shall check all images with respect to their expected checksums.				
Comments: There will be a final VM image for each component (ELEKTRA Core, ECMUX).				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-01287] Software Actualization - Initiation via DDSP	
Target Release:	VIL_OBB_1.1.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core		System test	
	ECMUX	See also:		
Issues/Decisions:				

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EL4-VIL-SSS-REQ-01308 VIL Self-Contained VM Images (Draft)	
The final VM images of VIL to be used for a certain station shall contain all required SW components and configuration data to perform its functions.	
<i>Comments:</i> This means that the VIL shall not depend on any services other than those provided through specified interfaces.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00217] VIL Redundancy Concept - Isolation of VIL Instances <i>Scope:</i> Sub-System <i>Verification:</i> Analysis <i>See also:</i>
<i>Issues/Decisions:</i>	

4.2 REDUNDANCY

EL4-VIL-SSS-REQ-00218 VIL Hot-Hot Redundancy (Basic) (Draft)	
In basic redundancy configuration, the VIL shall support up to two VIL redundant nodes running in hot-hot redundancy.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation - Redundant Nodes ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-00219		VIL Hot-Hot Redundancy (Georedundant)		(Draft)	
In georedundant configuration, the VIL shall support up to four VIL redundant nodes running in hot-hot redundancy. It shall be possible to run the nodes on two different locations (data centers), two nodes on each of them.					
Comments:					
Category: FUR		Priority: Mandatory		==> [EL4 SSS] [EL4-DIL-SSS-REQ-00457] DIL Variants - Geo-Redundancy - Different Servers ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00458] DIL Variants - Geo-Redundancy - Seamless Operation ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00456] DIL Variants - Geo-Redundancy - Different Data Center Locations	
Originator: Technics (R&D)		Origin:			
Target Release: VIL_OBB_2.0.0		Scope: Sub-System			
Allocation: VIL ELEKTRA Core ECMUX		Verification: Integration test			
		See also:			
Issues/Decisions:					

EL4-VIL-SSS-REQ-01446		VIL Switch RaSTA Transport Channels to Different Node		(In negotiation)
In georedundant configuration, after failure of a VIL redundant node with active RaSTA transport channels, the VIL shall be able to reestablish these transport channels on a different VIL redundant node.				
Comments: The trigger for channel switching (automatic or maintainer command) has to be agreed with the customer.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00991] VIL Redundancy Concept - Recovery	
Target Release:	VIL_OBB_2.0.0	Scope:	Sub-System	
Allocation:	VIL ELEKTRA Core ECMUX	Verification:	Integration test	
		See also:		
Issues/Decisions:				

EL4-VIL-SSS-REQ-00220		VIL Synchronized Startup of VIL Redundant Nodes		(Draft)	
The VIL redundant nodes (all or subset) shall be able to start operation in a synchronized way. All redundant nodes starting up within a certain time window shall enter operational mode simultaneously and without performing automatic reintegration ("recovery").					
Comments: An appropriate time window shall be defined during design based on practical experience.					
Category:	FUR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy		
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System		
Allocation:	VIL	Verification:	Integration test		
	ELEKTRA Core	See also:			
Issues/Decisions:					

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EL4-VIL-SSS-REQ-00221		VIL Automatic Reintegration		(Draft)	
When a VIL redundant node is started while its corresponding VIL instance is already operational, it shall be automatically integrated into the VIL instance without service interruption.					
Comments:					
Category:	FUR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection		
			==> [EL4 SSS] [EL4-DIL-SSS-REQ-00453] Software Actualization - General Request		
Target Release:	VIL_OBB_1.1.0	Scope:	Sub-System		
Allocation:	VIL	Verification:	Integration test		
	ELEKTRA Core	See also:			
	ECMUX				
Issues/Decisions:					

EL4-VIL-SSS-REQ-00222		VIL Split-Brain Prevention		(Draft)	
In case one or several VIL redundant nodes are disconnected from the other nodes, it shall be ensured that only one subgroup of synchronized nodes remains active.					
Comments: This avoids the split-brain problem, where redundant nodes are no longer synchronized and may send inconsistent control messages.					
Category:	FUR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy		
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System		
Allocation:	VIL	Verification:	Integration test		
	ELEKTRA Core	See also:			
	ECMUX				
Issues/Decisions:					

4.3 EXTERNAL INTERFACES

4.3.1 INTERFACES TO ADJACENT SYSTEMS

This chapter deals with interfaces to the following types of adjacent systems:

- OMC
- RBC
- SCWS
- Adjacent Interlocking
- Relay Block (via Interface Converter)

There are two kinds of interfaces to these adjacent systems to be implemented in different releases:

- X25-over-RaSTA
- Eulynx Interfaces (SCI-CC, SCI-ILS, SCI-RBC)

Please note that the interfaces of each type are based on generic functions that are described in section "Generic Functions" and contain the requirements that are common to all interfaces of the same type.

The requirements of this section deal only with interfaces themselves. The closely related generic function "Communicate via X25-over-RaSTA" is specified in section 4.5.3.

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4.3.1.1 OMC INTERFACE

The specification of the OMC interface can be found in the DIL system requirements specification [DIL_SSS].

4.3.1.2 SCWS INTERFACE

The specification of the SCWS interface can be found in the DIL system requirements specification [DIL_SSS].

4.3.1.3 RBC INTERFACE

The specification of the RBC interface can be found in the DIL system requirements specification [DIL_SSS].

4.3.1.4 ADJACENT IL INTERFACE

The specification of the Adjacent Interlocking interface can be found in the DIL system requirements specification [DIL_SSS].

4.3.1.5 INTERFACE TO RELAY BLOCK (VIA INTERFACE CONVERTER)

The specification of the interface to relay blocks via an Interface Converter can be found in the DIL system requirements specification [DIL_SSS].

4.3.2 FIELD ELEMENT INTERFACES

The VIL controls digital field elements by communicating with object controllers through Eulynx SCI interfaces of the following types:

- Point Interface (SCI-P)
- Signal Interface (SCI-LS)
- Generic I/O Interface (SCI-IO)
- Train Detection System Interface (SCI-TDS)
- Level Crossing Interface (SCI-LX)

The specification of these interfaces can be found in the DIL system requirements specification [DIL_SSS].

The requirements of this section deal only with the given interfaces. The closely related generic function "Communicate with Field Elements via SCI" is specified in section 4.5.4.

4.3.3 DIAGNOSIS INTERFACES

The VIL sends diagnosis data to an external diagnosis collector through different interfaces related to different data categories:

- Datalogger Interface: application-related diagnosis data (operational data, VIL status data) that is produced in a fully synchronised manner by the redundant nodes
- Syslog Interface: system diagnosis data (produced e.g. by TAS Platform libraries)
- VM Resource Monitoring Interface: infrastructure diagnosis data

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4.3.3.1 DATALOGGER INTERFACE

The requirements of this section deal only with the given interface between VIL and the external diagnosis collector. The closely related generic function "Provide Application Diagnosis Data" is described in section 4.5.5.

EL4-VIL-SSS-REQ-00287 Datalogger Interface Protocol		(Draft)
The VIL shall provide a Datalogger Interface for sending application diagnosis data. The interface shall be based on the RaSTA protocol and shall be implemented according to the detailed specifications in [DATLOGIF].		
<i>Comments:</i>		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0013] <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>	
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface Decision: ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.		

EL4-VIL-SSS-REQ-00288 Datalogger Interface - Basic Redundancy		(Draft)
In basic redundancy configuration, the VIL Datalogger Interface shall have one RaSTA connection, which shall have two redundant transport channels with endpoints on different VIL redundant nodes.		
<i>Comments:</i>		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation - Redundant Nodes <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>	
<i>Issues/Decisions:</i>		

EL4-VIL-SSS-REQ-01447 Datalogger Interface - Georedundancy		(In negotiation)
In georedundant configuration, the VIL Datalogger Interface shall support two alternative RaSTA connections to separate external diagnosis collector instances, one for each georedundant site. Each connection shall have two redundant transport channels with endpoints on different VIL redundant nodes of the same site. At any time at most one connection shall be established by the VIL. If the active connection fails, the VIL shall try to open the other one.		
<i>Comments:</i> This might be changed depending on the georedundancy concept of subsystem ADM.		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_2.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00456] DIL Variants - Geo-Redundancy - Different Data Center Locations <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>	
<i>Issues/Decisions:</i>		

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EL4-VIL-SSS-REQ-01307 Datalogger Interface - Message Integrity (Draft)	
The Datalogger Interface shall not use a safety code on the safety and retransmission layer of RaSTA. However it shall use a check code with 32-bit CRC on the redundancy layer.	
<i>Comments:</i> Since this is a non-safe connection, no MD4 safety code is needed. For the check code either option (b) or (c) as defined in [RASTA] can be selected during the design phase.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0013] <i>Scope:</i> Sub-System <i>Verification:</i> Code reading Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface Decision: ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.	

EL4-VIL-SSS-REQ-00289 Datalogger Interface - RaSTA Configuration (Draft)	
The VIL shall support station-specific configuration of the datalogger interface RaSTA connection, including at least the following parameters: Safety layer: • sender identification • receiver identification Transport layer (per transport channel): • local IP address & UDP port • remote IP address & UDP port	
<i>Comments:</i> The check code computation has a fixed initial value and therefore does not require station-specific configuration.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0013] <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface Decision: ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.	

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EL4-VIL-SSS-REQ-00291 Datalogger Interface - Header Data		(Draft)
All diagnosis message types shall have a common section including at least the following data:		
• message type identifier • message origin (application that generated the message) • sequence number • timestamp (in UTC)		
<i>Comments:</i> The sequence number allows unambiguous mapping of each acknowledgement message to the corresponding diagnosis message.		
Category:	FUR	Priority: Mandatory
Originator:	Technics (R&D)	Origin: ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0013]
Target Release:	VIL_OBB_1.0.0	Scope: Sub-System
Allocation:	VIL ELEKTRA Core	Verification: Code reading Module test
<i>See also:</i>		
Issues/Decisions:	EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface Decision: ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.	

EL4-VIL-SSS-REQ-00292 Datalogger Interface - Human-Readable Identifiers		(Draft)
Station-specific object identifiers (e.g. station, field elements) contained in diagnostic messages shall be human-readable (i.e. meaningful alpha-numeric strings as defined by data preparation).		
<i>Comments:</i> This frees the diagnosis collector from mapping internal VIL identifiers.		
Category:	FUR	Priority: Mandatory
Originator:	Technics (R&D)	Origin: ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0013]
Target Release:	VIL_OBB_1.0.0	Scope: Sub-System
Allocation:	VIL ELEKTRA Core	Verification: Integration test
<i>See also:</i>		
Issues/Decisions:	EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface Decision: ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.	

EL4-VIL-SSS-REQ-00293 Datalogger Interface - Message Transmission		(Draft)
Whenever the RaSTA connection is established, the VIL shall sequentially send diagnosis messages in the order of generation (oldest message first). After sending a message the VIL shall wait for the reception of the corresponding acknowledgement message before continuing with the next message.		
<i>Comments:</i>		
Category:	FUR	Priority: Mandatory
Originator:	Technics (R&D)	Origin: ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0013]
Target Release:	VIL_OBB_1.0.0	Scope: Sub-System
Allocation:	VIL ELEKTRA Core	Verification: Integration test Module test
<i>See also:</i>		
Issues/Decisions:	EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface Decision: ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.	

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4.3.3.2 SYSLOG INTERFACE

EL4-VIL-SSS-REQ-01448		VIL Syslog Interface - Configurable Destination		(Draft)
The VIL shall send technical diagnosis data produced by its SW components to the external diagnosis collector. For this purpose every VM of the VIL shall establish a dedicated TCP connection to a syslog server. The IP address and port of the syslog server shall be configurable by station-specific configuration.				
Comments: The log messages sent via the Syslog interface are not replica-deterministic. The syslog messages will be sent via a bonded network. Therefore redundant transmission by VIL is not required.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation - Redundant Nodes	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL ELEKTRA Core ECMUX	Verification:	Integration test	
		See also:		
Issues/Decisions:				

EL4-VIL-SSS-REQ-01449		VIL Syslog Interface - Destination per Site		(In negotiation)
In georedundant configuration, the VIL shall be configurable to send the syslog data of the VMs to separate syslog servers on each georedundant site. The IP addresses of both syslog servers shall be configurable by station-specific configuration.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00456] DIL Variants - Geo-Redundancy - Different Data Center Locations	
Target Release:	VIL_OBB_2.0.0	Scope:	Sub-System	
Allocation:	VIL ELEKTRA Core ECMUX	Verification:	Integration test	
		See also:		
Issues/Decisions:				

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4.3.3.3 VM RESOURCE MONITORING INTERFACE

EL4-VIL-SSS-REQ-01144 VM Resource Management Interface - Data Categories (Draft)	
Each Virtual Machine belonging to the VIL shall provide an interface for monitoring the resource status of the VM's guest operating system. This interface shall provide at least the following data: • VM universal identifier • VM system time • CPU utilization (percent) • Memory usage (used vs. total available) • Virtual disk space (used vs. total available) • VLAN network traffic (outgoing, incoming) • Running processes (incl. CPU and memory usage) Note that these data shall be provided for each VM of each active VIL redundant node.	
<i>Comments:</i> The corresponding communication protocol (e.g. SNMP) has still to be defined.	
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0014] <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0014 VIL VM Resource Management Interface Decision: The VIL shall provide an interface for each of its virtual machines in order to monitor the VM's resource status.	

4.3.4 SECURITY LOGGING INTERFACE

The VIL security logging interface will be specified in [VIL_SSS_SEC].

4.3.5 TIME SYNCHRONISATION INTERFACE

EL4-VIL-SSS-REQ-00232 VIL Time Synchronisation Interface (Draft)	
All Virtual Machines belonging to VIL shall synchronise their system time with the corresponding host server's system time using the virtual PTP hardware clock provided by the KVM hypervisor.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00557] Time Synchronisation - NTP Time Server - Usage <i>Scope:</i> Sub-System <i>Verification:</i> Code reading Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

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4.3.6 APPLICATION MAINTAINER INTERFACE

EL4-VIL-SSS-REQ-01450 Application Maintainer Interface for CHILL Applications		(In negotiation)
The VIL shall provide an Application Maintainer Interface allowing to check detailed SW states at runtime in order to analyse application SW problems of the CHILL applications. For this purpose each running VIL redundant node shall provide at least one TCP connection per application that can be opened at runtime by a client.		
<i>Comments:</i> The interface can be used as contribution to the Eulynx Maintainer Interface (EIL4).		
<i>Category:</i>	FUR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00482] Maintainer - Interface EIL4
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i> Sub-System
<i>Allocation:</i>	VIL ELEKTRA Core	<i>Verification:</i> Integration test
<i>See also:</i>		
<i>Issues/Decisions:</i>		

4.3.7 TECHNICAL INTERFACES (TO IPS)

4.3.7.1 PKI INTERFACE

The PKI interface of subsystem VIL will be specified in [VIL_SSS_SEC].

4.3.7.2 OBJECT STORAGE INTERFACE

EL4-VIL-SSS-REQ-01451 Object Storage Interface		(Draft)
The VIL shall be able to store, retrieve and delete status data in/from a persistent object storage. The VIL shall access the object storage service using the http protocol. The URL of the object storage service shall be configurable by station-specific configuration.		
<i>Comments:</i> (1) Open point: use of https (2) Open point: how to separate data of different stations (URL or client certificate)		
<i>Category:</i>	FUR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00662] Startup all VIL Instances - Configuration Change - Necessary Data after long-term Shutdown
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i> Sub-System
<i>Allocation:</i>	VIL ELEKTRA Core	<i>Verification:</i> Integration test
<i>See also:</i>		
<i>Issues/Decisions:</i>		

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4.3.7.3 CDS INTERFACE

EL4-VIL-SSS-REQ-01452 Cluster Deployment Supervisor Interface (Draft)	
Each VM of the VIL shall communicate with the Cluster Deployment Supervisor (CDS) through the CDS interface. The IP address of the CDS shall be configurable by station-specific configuration.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01453 Cluster Deployment Supervisor Interface - Georedundancy (Draft)	
In georedundant configuration it shall be possible to configure separate IP addresses of the Cluster Deployment Supervisor on each site. Each VM shall use only the IP address configured for its site.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

4.3.7.4 VM DEPLOYMENT DESCRIPTION

EL4-VIL-SSS-REQ-01454 VM Deployment Description (In negotiation)	
The VIL instance deliverable shall include a deployment description file specifying the resources that the VIL instance requires from the IPS cluster management. It shall include at least: • Redundancy topology • For each VM: networks to be connected with virtual network devices • For each VM: required processing resources (memory, CPU) and minimum performance values	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00437] Architecture - Number of VIL Instances per Server <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

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4.4 OPERATIONAL FUNCTIONS

4.4.1 CONTROL TRAIN MOVEMENTS

The detailed specification of the operational function "Control Train Movements" can be found in following sections of the System SSS [DIL SSS]:

- Routes
- Flank Protection
- Opposite Run Protection
- Line Protection
- Local Control
- Universal Element
- Train Movement Lock

EL4-VIL-SSS-REQ-01455 Control Train Movements - Safety Integrity Level (Draft)			
Function "Control Train Movements" shall be developed as a SIL-4 function according to [EN-50716].			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00686] Safety - Integrity Levels - Development DIL
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL ELEKTRA Core ECMUX	<i>Verification:</i>	Review
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

4.4.2 CONTROL FIELD ELEMENTS

The detailed specification of the operational function "Control Field Elements" can be found in following sections of the System SSS [DIL_SSS_CU1]:

- Central Functions Eulynx Subsystem Point
- Central Functions Eulynx Subsystem External LCSi
- Central Functions Eulynx Subsystem Light Signal
- Central Functions Eulynx Subsystem Generic Input/Output
- Central Functions Eulynx Subsystem Train Detection System

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EL4-VIL-SSS-REQ-01456		Control Field Elements - Safety Integrity Level		(Draft)
Function "Control Field Elements" shall be developed as a SIL-4 function according to [EN-50716].				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00686] Safety - Integrity Levels - Development DIL	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Review	
	ELEKTRA Core	See also:		
	ECMUX			
Issues/Decisions:				

4.4.3 SEND OPERATOR WARNINGS

The detailed specification of the operational function "Send Operator Warnings" can be found in following section of the System SSS [DIL SSS]:

- Summation Messages

4.5 GENERIC FUNCTIONS

4.5.1 TIME SYNCHRONISATION

EL4-VIL-SSS-REQ-00233		VIL Time Synchronisation Diagnosis Data		(Draft)
The VIL shall send diagnosis data containing the time synchronisation status of the given VIL instance. The data shall be sent to the external diagnosis collector through the Datalogger Interface at least upon startup and each status change.				
Comments: The status shall be the same as reported in the "summation disturbance notification" (X25 message group 48) to OMC.				
Category:	MDR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:		
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Code reading	
	ELEKTRA Core		Integration test	
			Module test	
		See also:	==> [EL4-VIL-SSS-REQ-00283] Status-Related Diagnosis Messages	
Issues/Decisions:				

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4.5.2 GENERIC COMMUNICATION REQUIREMENTS

4.5.2.1 NETWORK LAYER CONFIGURATION

EL4-VIL-SSS-REQ-00281 VIL Communication - IP Configuration		(Draft)
For each network interface to be used by one of its virtual machines the VIL shall allow configuring • local IP address and netmask • network interface name For each destination network that is only reachable via an IP gateway, the following route parameters shall be configurable: • destination network (IP address and netmask) • gateway IP address		
<i>Comments:</i> The term "network interface" refers to the virtual Ethernet devices to be provided to the VIL's virtual machines.		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Review Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00286] RaSTA Communication - Automatically Open Connections	
<i>Issues/Decisions:</i>		

4.5.2.2 GENERIC RASTA CONNECTION REQUIREMENTS

EL4-VIL-SSS-REQ-00286 RaSTA Communication - Automatically Open Connections		(Draft)
Upon startup the VIL shall automatically try to establish all configured RaSTA connections (as a server or client) according to the specifications in [RASTA]. After a connection loss it shall periodically try to reopen the connection.		
<i>Comments:</i> (1) The connections of the Datalogger Interface are an exception to this requirement (see EL4-VIL-SSS-REQ-01447). (2) There may be situations where a connection shall not be automatically reopened due to security requirements specified in [VIL_SSS_SEC].		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00235] Generic Function for X25-over-RaSTA Communication ==> [EL4-VIL-SSS-REQ-00709] Generic Function Communicate with FE via SCI	
<i>Issues/Decisions:</i>		

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EL4-VIL-SSS-REQ-00244 RaSTA Communication - Connection Status (Draft)	
For each external RaSTA connection, VIL shall send diagnosis data containing the connection status. The data shall be sent through the Datalogger Interface to the external diagnosis collector at least upon startup and each status change.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00329] Reliability - Diagnosis of Failure of open Network ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00330] Reliability - Diagnosis of failure of closed Network <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00283] Status-Related Diagnosis Messages
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00245 RaSTA Communication - Transport Channel Status (Draft)	
For each external RaSTA connection, VIL shall send diagnosis data containing the transport channel status of each associated RaSTA transport channel. The data shall be sent through the Datalogger Interface to the external diagnosis collector at least upon startup and each status change.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00329] Reliability - Diagnosis of Failure of open Network ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00330] Reliability - Diagnosis of failure of closed Network <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00283] Status-Related Diagnosis Messages
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00246 RaSTA Communication - Connection Statistics (Draft)	
For each external RaSTA connection, VIL shall send diagnosis data containing error counter statistics. The data shall be sent through the Datalogger Interface to the external diagnosis collector every 24 hours. After reading the statistics the counters shall be reset. The transmission of the diagnosis data shall start at 01:00 a.m. UTC with 2 min. tolerance.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00329] Reliability - Diagnosis of Failure of open Network ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00330] Reliability - Diagnosis of failure of closed Network <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-00279		RaSTA Communication - Notify Connection Loss to OMC		(Draft)
The VIL has to notify the controlling OMC about any loss of a RaSTA connection as a system error ("Anlagenfehler"). The system error shall be cancelled only when no failure is pending again. The notification shall be sent through X25 message 48 (bit "Anlagenfehler") according to the rules given in [PH_X25].				
Comments: Failure of a redundant transport channel (while the connection remains open) shall not lead to such notification.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00988] Summation Messages - Reference	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL ELEKTRA Core	Verification:	Integration test Module test	
		See also:		
Issues/Decisions:				

The management of encryption keys and certificates (for TLS communication) within VIL shall be specified in [VIL_SSS_REC].

4.5.3 GENERIC FUNCTION "COMMUNICATE VIA X25-OVER-RASTA"

EL4-VIL-SSS-REQ-00235		Generic Function for X25-over-RaSTA Communication		(Draft)
The VIL shall provide a generic function "Communicate via X25-over-RaSTA" that supports X25-over-RaSTA interfaces to the adjacent systems. These interfaces shall be compliant with the specifications in [X25-over-IP] and [RASTA].				
Comments: The complete X25 telegrams shall be embedded into the RaSTA layer.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00464] Interface to OMC - X25 over RaSTA ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00468] Interface to RBC - Communication Protocol ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00470] Interface to SCWS - Communication Protocol ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00732] Interface to Adjacent IL - Other Adjacent IL - Communication Protocol	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL ELEKTRA Core	Verification:	Analysis Integration test	
		See also:		
Issues/Decisions:				

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EL4-VIL-SSS-REQ-00274 Support X25 Protocol		(Draft)
The VIL shall support the X25 protocol as specified in [PH_X25] with the following exceptions: - Requirements regarding data transmission via HDLC are not applicable. - Requirements regarding redundant X25 connections (primary/secondary switch-over) are not applicable.		
<i>Comments:</i> VIL provides redundancy for availability using the redundancy layer of the RaSTA protocol. Therefore primary/secondary switch-over is not necessary.		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00464] Interface to OMC - X25 over RaSTA ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00468] Interface to RBC - Communication Protocol ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00470] Interface to SCWS - Communication Protocol ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00732] Interface to Adjacent IL - Other Adjacent IL - Communication Protocol	
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test	
<i>See also:</i>		
<i>Issues/Decisions:</i>		

EL4-VIL-SSS-REQ-01309 X25-over-RaSTA Support TLS Connections		(Draft)
The VIL shall support X25-over-RaSTA interfaces based on UDP or TLS on transport layer. The protocol to be used (UDP or TLS) shall be configurable by station-specific configuration for each individual X25 communication partner.		
<i>Comments:</i>		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test	
<i>See also:</i>		
<i>Issues/Decisions:</i>		

EL4-VIL-SSS-REQ-00238 X25-over-RaSTA Safe Connections		(Draft)
The VIL shall support safe X25-over-RaSTA interfaces, which shall be realized by two RaSTA connections, one transmitting non-inverted messages and one transmitting inverted messages.		
<i>Comments:</i>		
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00464] Interface to OMC - X25 over RaSTA ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00468] Interface to RBC - Communication Protocol ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00470] Interface to SCWS - Communication Protocol ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00732] Interface to Adjacent IL - Other Adjacent IL - Communication Protocol	
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL	<i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test	
<i>See also:</i>		
<i>Issues/Decisions:</i>		

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EL4-VIL-SSS-REQ-00239 X25-over-RaSTA Non-safe Connections (Descoped)	
The VIL shall support non-safe X25-over-RaSTA interfaces, which shall be realized by one RaSTA connection transmitting non-inverted messages.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00247 X25-over-RaSTA Station-Specific Connection Parameters (Draft)	
For each X25-over-RaSTA connection, the VIL shall support station-specific configuration of the following parameters: Transport layer parameters (per transport channel): • Local IP address and port • Remote IP address and port Safety and retransmission layer parameters (per RaSTA connection): • Initial value of MD4 safety code • Sender ID • Receiver ID	
<i>Comments:</i> The remote IP address is not required, if on transport layer TLS is used and the VIL is server.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00235] Generic Function for X25-over-RaSTA Communication
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00251 X25-over-RaSTA Safety Integrity Level (Draft)	
Function "Communicate via X25-over-RaSTA" shall be developed as a SIL-4 function according to [EN-50716].	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00683] Safety - Integrity Levels - Interface - Adjacent Systems <i>Scope:</i> Sub-System <i>Verification:</i> Review <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-00275 X25-over-RaSTA Input Voting		(Draft)
For each safe X25-over-RaSTA interface, the VIL shall perform input voting of the non-inverted and inverted X25 channels before forwarding the received data to the X25 layer. In case of a voting error the X25 connection shall be closed and reopened again.		
<i>Comments:</i> Output voting is not required as each output message has to be voted by the communication partner.		
Category:	FUR	Priority: Mandatory
Originator:	Technics (R&D)	Origin:
Target Release:	VIL_OBB_1.0.0	Scope: Sub-System
Allocation:	VIL	Verification: Module test
	ELEKTRA Core	See also: ==> [EL4-VIL-SSS-REQ-00235] Generic Function for X25-over-RaSTA Communication
<i>Issues/Decisions:</i>		

EL4-VIL-SSS-REQ-00276 X25-over-RaSTA Safety Reaction		(Draft)
Whenever function "Communicate via X25-over-RaSTA" cannot guarantee valid transmission of commands and notifications (e.g. due to failure of internal communication), the VIL shall reboot the SW components controlling the affected X25-over-RaSTA connections.		
<i>Comments:</i> The reboot will lead to disconnection of the underlying RaSTA connections.		
Category:	FUR	Priority: Mandatory
Originator:	Technics (R&D)	Origin:
Target Release:	VIL_OBB_1.0.0	Scope: Sub-System
Allocation:	VIL	Verification: Integration test
	ELEKTRA Core	See also: ==> [EL4-VIL-SSS-REQ-00251] X25-over-RaSTA Safety Integrity Level
<i>Issues/Decisions:</i>		

EL4-VIL-SSS-REQ-00277 X25-over-RaSTA Transport Channel Redundancy		(Draft)
For each X25-over-RaSTA interface, the VIL shall support two transport channels (on RaSTA redundancy layer) via different virtual networks.		
<i>Comments:</i> This provides robustness against network failure.		
Category:	FUR	Priority: Mandatory
Originator:	Technics (R&D)	Origin:
Target Release:	VIL_OBB_1.0.0	Scope: Sub-System
Allocation:	VIL	Verification: Review
	ELEKTRA Core	Integration test
		See also: ==> [EL4-VIL-SSS-REQ-00235] Generic Function for X25-over-RaSTA Communication
<i>Issues/Decisions:</i>		

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EL4-VIL-SSS-REQ-00278 X25-over-RaSTA Transport Channel Separation (Draft)			
For each X25-over-RaSTA interface, the VIL shall support redundant transport channels with endpoints on different redundant nodes.			
<i>Comments:</i> This provides robustness against server failure.			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00699] DIL Variants - Geo-Redundancy - Splitting of Transport Channels
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL ELEKTRA Core	<i>Verification:</i>	Review Integration test
<i>See also:</i>			
<i>Issues/Decisions:</i>			

EL4-VIL-SSS-REQ-00280 X25-over-RaSTA Connection Status (X25 layer) (Draft)			
For each X25-over-RaSTA interface, VIL shall report the connection status of the X25 connection via the Datalogger Interface to the external diagnosis collector at least upon startup and each status change.			
<i>Comments:</i>			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL ELEKTRA Core	<i>Verification:</i>	Integration test Module test
<i>See also:</i> ==> [EL4-VIL-SSS-REQ-00283] Status-Related Diagnosis Messages			
<i>Issues/Decisions:</i>			

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4.5.4 GENERIC FUNCTION "COMMUNICATE WITH FIELD ELEMENTS VIA SCI"

EL4-VIL-SSS-REQ-00709 Generic Function Communicate with FE via SCI (Draft)			
The VIL shall provide a generic function "Communicate with Field Elements via SCI" that supports EULYNX SCI interfaces to field elements according to the following EULYNX specification (Baseline Set 4 Release 4): Interface specification SCI Generic [Eu.Doc.93] On transport layer TLS shall be used.			
<i>Comments:</i>			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00612] Interfaces - Point - SCI-P ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00615] Interface SCI-LX - Communication ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00483] Interfaces - Light Signals - SCI-LS ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00502] Interfaces - SCI-IO ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00503] Interfaces - Train Detection System - SCI-TDS
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL ECMUX	<i>Verification:</i>	Analysis Module test Subit test
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

EL4-VIL-SSS-REQ-00710 Communicate with FE via SCI - Safety Integrity Level (Draft)			
Function "Communicate with Field Elements via SCI" shall be developed as a SIL-4 function according to [EN-50716].			
<i>Comments:</i>			
<i>Category:</i>	SAR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00684] Safety - Integrity Levels - Interface - OC
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL ECMUX	<i>Verification:</i>	Analysis
		<i>See also:</i>	
<i>Issues/Decisions:</i>			

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EL4-VIL-SSS-REQ-00711 Communicate with FE via SCI - Station-Specific Connection Parameters (Draft)	
For each SCI connection to a field element, the VIL shall support station-specific configuration of the following parameters: Transport layer parameters (per transport channel): • Local IP address and port • Remote IP address and port Safety and retransmission layer parameters (per RaSTA connection): • Initial value of MD4 safety code • Sender ID • Receiver ID	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ECMUX CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00709] Generic Function Communicate with FE via SCI
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00712 Communicate with FE via SCI - Safety Reaction (Draft)	
Whenever function "Communicate with Field Elements via SCI" cannot guarantee valid transmission of commands and notifications (e.g. due to failure of internal communication), the VIL shall set all affected Field Elements to a safe state by sending the corresponding safe state commands to their Object Controllers.	
<i>Comments:</i> Here "failure of internal communication" refers to the ECMUX Command and Notification Interface (between the CC and ECMUX applications).	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00710] Communicate with FE via SCI - Safety Integrity Level
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-00713 Communicate with FE via SCI - Transport Channel Redundancy (Draft)	
For each field element interface via SCI, the VIL shall support two transport channels (on RaSTA redundancy layer) via different virtual networks.	
<i>Comments:</i> This provides robustness against network failure.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Review Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00709] Generic Function Communicate with FE via SCI
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00714 Communicate with FE via SCI - Transport Channel Separation (Draft)	
For each field element interface via SCI, the VIL shall support redundant transport channels with endpoints on different redundant nodes.	
<i>Comments:</i> This provides robustness against server failure.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00699] DIL Variants - Geo-Redundancy - Splitting of Transport Channels <i>Scope:</i> Sub-System <i>Verification:</i> Review Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00715 Communicate with FE via SCI - Connection Status (PDI layer) (Draft)	
For each field element interface via SCI, the VIL shall report the connection status on application layer (PDI protocol) to the external diagnosis collector at least upon startup and each status change.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-00283] Status-Related Diagnosis Messages
<i>Issues/Decisions:</i>	

4.5.5 GENERIC FUNCTION "PROVIDE APPLICATION DIAGNOSIS DATA"

The task of VIL function "Provide Application Diagnosis Data" is to collect internal diagnosis data from its SW applications (e.g. CC, ECMUX, RCU, DGP), serialize it and send it through the Datalogger Interface.

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EL4-VIL-SSS-REQ-00282 Convert and Send Diagnosis Messages (Draft)	
Whenever VIL function "Provide Application Diagnosis Data" receives diagnosis data from an internal SW application, it shall convert the data to the corresponding diagnosis message and send it via the Datalogger Interface to the external diagnosis collector.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00283 Status-Related Diagnosis Messages (Draft)	
After connecting or reconnecting to the external diagnosis collector the VIL shall send a set of status-related diagnosis messages covering at least the following status data:	
• Version data (SW, configuration data, station data) • Application status (including "hex display" data) • VIL redundancy status • VIL time synchronisation status • Connection status of external interfaces (including RaSTA connection and transport channel status) • Connection status of internal interfaces (including RaSTA connection and transport channel status)	
<i>Comments:</i> Note that "Connection status of external interfaces" also refers to the Datalogger Interface.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL <i>Scope:</i> Sub-System <i>Verification:</i> Integration test Module test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00284 VIL Diagnosis Processing - Safety Integrity Level (Draft)	
Function "Provide Application Diagnosis Data" shall be developed as "Basic Integrity" function according to [EN-50716].	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00685] Safety - Integrity Levels - Interface - DDSP, PS <i>Scope:</i> Sub-System <i>Verification:</i> Analysis <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-00285		VIL Diagnosis Processing - Redundancy		(Draft)
Function "Provide Application Diagnosis Data" shall be available whenever at least one VIL redundant node is running. For this purpose it shall run fully synchronized on all redundant nodes (hot-hot redundancy).				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core	See also:		
Issues/Decisions:				

EL4-VIL-SSS-REQ-00294		Diagnosis Buffer		(Draft)
Function "Provide Application Diagnosis Data" shall store every diagnosis message immediately after its generation together with the current timestamp in a buffer. Each message shall remain stored in the buffer until it has been sent and the corresponding acknowledgement message with the same sequence number has been received.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core	See also:	Module test	
Issues/Decisions:				

EL4-VIL-SSS-REQ-00295		Diagnosis Message Response Timeout		(Draft)
The VIL shall allow configuration of a diagnosis response timeout. The corresponding timer shall be started upon sending of a diagnosis message. If the timer expires before the expected acknowledgement is received, the diagnosis connection shall be closed and reopened.				
Comments: The purpose of the timeout is to avoid a deadlock after failure of the external diagnosis collector.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core		Module test	
		See also:		
Issues/Decisions:				

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EL4-VIL-SSS-REQ-00296		Diagnosis Buffer Capacity		(Draft)	
The diagnosis buffer shall be able to store up to 15000 diagnosis messages. In case of overflow the oldest diagnosis data shall be overwritten first. An additional diagnosis message shall be inserted to indicate the loss of diagnosis data.					
Comments: The given value is based on the ELEKTRA 2.0 implementation.					
Category:	FUR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL		
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System		
Allocation:	VIL	Verification:	Module test		
	ELEKTRA Core	See also:			
Issues/Decisions:					

EL4-VIL-SSS-REQ-00297		Diagnosis Buffer Persistency		(Draft)	
In case of a controlled shutdown of the SW component storing the diagnosis buffer, the VIL shall dump the diagnosis buffer to persistent storage. Upon restart of that component the dumped diagnosis buffer shall be reloaded and the contained diagnosis messages shall be transmitted to the diagnosis collector.					
Comments: In the case of a hard shutdown a potential loss of diagnosis messages is acceptable.					
Category:	FUR	Priority:	Mandatory		
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL		
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System		
Allocation:	VIL	Verification:	Integration test		
	ELEKTRA Core	See also:			
Issues/Decisions:					

4.6 GENERAL REQUIREMENTS

EL4-VIL-SSS-REQ-01143		VIL Robustness to Year 2038 Problem		(Draft)	
The VIL functions shall not be affected by time representation problems before year 2100. In particular they shall not be disturbed by the so-called "Year 2038 Problem" (overflow of UTC seconds on 19.01.2038 when represented by a signed 32-bit integer).					
Comments: See https://en.wikipedia.org/wiki/Year_2038_problem for a more detailed explanation.					
Category:		FUR	Priority:		Mandatory
Originator:		Technics (R&D)	Origin:		
Target Release:		VIL_OBB_1.0.0	Scope:		Sub-System
Allocation:		VIL	Verification:		Review
		ELEKTRA Core			Integration test
		ECMUX	See also:		
Issues/Decisions:					

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4.6.1 SHUTDOWN AND RESTART

EL4-VIL-SSS-REQ-01155 VIL Application Restart (Draft)	
In the event of a SW failure of one of the VIL software applications, the VIL shall automatically stop and restart the affected application on all redundant nodes. If the application consists of two diverse SW channels, both channels shall be stopped and restarted.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01156 VIL Maintenance Shutdown (Draft)	
The VIL shall provide a way to trigger a controlled maintenance shutdown of the complete VIL instance, consisting of • graceful shutdown of all SW applications • subsequent OS shutdown on all Virtual Machines of all VIL redundant nodes. In case of a maintenance shutdown the VIL shall not automatically reboot.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01294] Software Actualization - VIL - Serial Activation <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01457 VIL Maintenance Shutdown – Activation Command from OMC (In negotiation)	
The VIL shall support requesting and receiving an activation command to be sent by OMC before activating a VIL maintenance shutdown. This includes • Sending a notification to OMC (X25 notification: Meldung 57 Fernbedienung / Steuerwort 1 / Bit 4) • Waiting for the activation command from OMC (X25 command: Kommando 177 Fernbedienung / Steuerwort 1 / Bit 6)	
<i>Comments:</i> Open point: The corresponding interaction between DDSP, ADM and VIL is still to be defined in detail.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.1.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00695] Software Actualization - Activation Command from OMC - New Version <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

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4.6.2 PERSISTENT STATUS DATA

EL4-VIL-SSS-REQ-01157 Persistent Status Data - Local Storage (Draft)	
The VIL shall not lose its persistent status data in case of an application restart. For this purpose every VIL application maintaining persistent status data shall perform the following actions: • Before application exit the application shall store its persistent status data on the local temporary file system of the VM by writing a data file and a corresponding safety code file. • After application restart the application shall load the persistent status data from the local file, check the safety code and (if check was successful) restore the status data. It shall be ensured that a specific persistent status data file, once loaded by the application, cannot be loaded a second time.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00658] Startup all VIL Nodes - No Configuration Change - Necessary Data after short-term Shutdown - General
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01158 Persistent Status Data - Remote Storage (Draft)	
The VIL shall not lose its persistent status data even in case of a complete VIL shutdown and subsequent reboot. For this purpose the VIL shall perform the following actions: • During shutdown: After application shutdown, exactly one VIL redundant node shall upload its locally stored persistent status files to a configurable remote object storage. • During reboot: Before starting the applications the VIL shall download the remotely stored files to the temporary file system of the corresponding VM.	
<i>Comments:</i> After shutdown of a VM any data stored on the local temporary file system is lost.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00662] Startup all VIL Instances - Configuration Change - Necessary Data after long-term Shutdown
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-01466 Persistent Status Data - Prevent Download of Obsolete Data (Draft)	
The VIL shall ensure that a stored persistent status data file, once downloaded during a successful VIL reboot, cannot be downloaded again during a second VIL reboot. For this purpose the VIL shall delete the files as soon as it has reached operational state.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00662] Startup all VIL Instances - Configuration Change - Necessary Data after long-term Shutdown
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01180 Persistent Status Data - Check Data Source (Draft)	
Upon application restart each VIL application maintaining persistent status data shall check that the persistent status data file to be loaded was written by the same application and belongs to the same VIL instance. The check shall be based on application name and station ID. If the check fails, the VIL shall reject the file and continue as defined in EL4-VIL-SSS-REQ-01181.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00662] Startup all VIL Instances - Configuration Change - Necessary Data after long-term Shutdown
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01160 Persistent Status Data - Check Compatibility with Application Version (Draft)	
The VIL shall load a persistent status data file even after an application SW update, if the data format of the file is compatible with the new application version. Therefore, each VIL application maintaining persistent status data shall check upon application restart that the persistent status data file is compatible with the current application version. The check shall be based on a suitable database schema version check. If the check fails, the VIL application shall reject the persistent status data file and continue as defined in EL4-VIL-SSS-REQ-01181.	
<i>Comments:</i> "Compatibility with application version" means that the persistent status data has exactly the same size, structure, data types and meaning as for the currently running application version.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D)	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00731] Startup all VIL Instances - Configuration Change - Default Values after long-term Shutdown - General
<i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core	<i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-01499		Persistent Status Data - Check Compatibility with Station Data		(Draft)
Each VIL application maintaining persistent status data shall check the compatibility of the persistent status data file to be loaded with the current station-specific configuration data (station data) of that application. The check shall be based on a suitable station data version check. The persistent status data shall only be used in case of unchanged station-specific configuration data or in case of minor changes that do not affect validity and completeness of the persistent status data. In case of a major change (e.g. new element added) the VIL application shall reject the persistent status data file and continue as defined in EL4-VIL-SSS-REQ-01181.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00731] Startup all VIL Instances - Configuration Change - Default Values after long-term Shutdown - General	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core	See also:		
Issues/Decisions:				

EL4-VIL-SSS-REQ-01181		Persistent Status Data - Missing, Corrupted or Incompatible File		(Draft)
If any VIL persistent status data cannot be loaded successfully, e.g. because a file is missing, corrupted or incompatible with the current version, the VIL shall initialize all affected status parameters with the corresponding “default values after long downtime” as defined in EL4-DIL-SSS-REQ-00731 and generate a suitable diagnosis message.				
Comments: The “default values after long downtime” are identical to the initial values upon first VIL startup.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00662] Startup all VIL Instances - Configuration Change - Necessary Data after long-term Shutdown ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00731] Startup all VIL Instances - Configuration Change - Default Values after long-term Shutdown - General	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core	See also:		
Issues/Decisions:				

EL4-VIL-SSS-REQ-01501		Persistent Status Data - Report Data Loss to OMC		(Draft)
The VIL shall report any loss of persistent status data upon VIL or application restart to the OMC using X25 message 124 as defined in requirement FUR:6731.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00669] Recommissioning Command - Restore State	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	VIL	Verification:	Integration test	
	ELEKTRA Core	See also:		
Issues/Decisions:				

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4.6.3 INTERNAL CONSISTENCY

EL4-VIL-SSS-REQ-01458 Request authorization by CDS (Draft)	
Every VM of the VIL shall request upon startup an "authorization to execute" from the Cluster Deployment Supervisor (CDS). The request parameters shall be unique for each VM foreseen in the cluster. Therefore at least the following data shall be encoded in the request parameters: • Station id • VIL component (ELEKTRA Core or ECMUX) • VM id (ELEKTRA Core: CN and CE, ECMUX: CN) In addition, the request parameters shall also include the following version-related data: • Image version • Image checksum The VM shall not become active until a valid authorization is received. After successful startup it shall periodically check the authorization. In case of rejection the VM shall perform a permanent shutdown.	
<i>Comments:</i> (1) Based on this requirement, the CDS mechanism prevents operation of invalid VMs that are duplicate or have a wrong SW version. (2) The CDS is a SW component of TAS Platform Core.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01290] Software Actualization - Distribution - Check of Correct Versions <i>Scope:</i> Sub-System <i>Verification:</i> <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01459 Application SW Versions - Check Consistency between Nodes (Draft)	
The VIL shall detect any inconsistency between the SW versions of safety-critical applications (including configuration data) on different redundant nodes of a VIL instance. In case of inconsistency it shall be ensured that only nodes with consistent SW versions become operational.	
<i>Comments:</i> This requirement can be satisfied using the communication with the Cluster Deployment Supervisor (CDS).	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01290] Software Actualization - Distribution - Check of Correct Versions <i>Scope:</i> Sub-System <i>Verification:</i> <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-01460 Application SW Versions - Check Compatibility (Draft)	
The VIL shall check upon startup that the versions of its internal SW applications are compatible with each other. If an incompatible version is detected, the VIL shall perform a permanent shutdown.	
<i>Comments:</i> This requirement can be satisfied using the communication with the Cluster Deployment Supervisor (CDS).	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> <i>See also:</i>
<i>Issues/Decisions:</i>	

4.6.4 REACTIVITY MONITORING

EL4-VIL-SSS-REQ-01461 Reactivity Monitoring between Applications (Draft)	
The VIL shall periodically check the reactivity of its SW applications by monitoring the message response times between the applications. The maximum response time shall be configurable per application. In case a limit is exceeded, a suitable safety reaction shall be executed depending on the affected application.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> <i>See also:</i>
<i>Issues/Decisions:</i>	

4.6.5 FAILURE DETECTION TIMES

EL4-VIL-SSS-REQ-01462 Maximum Failure Detection Time - Internal (In negotiation)	
The VIL shall detect any internal HW failure, SW failure or internal connection failure and execute an appropriate reaction within [TBD] seconds.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> <i>See also:</i>
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-01463		Maximum Failure Detection Time – SCI Connections to Adjacent Systems	(In negotiation)
The VIL shall detect any failure of an SCI connection to an adjacent system and perform an appropriate reaction within [TBD] seconds.			
<i>Comments:</i> The value depends on the interface timeout defined in the X25 and SCI interface specifications.			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL	<i>Verification:</i>	
	ELEKTRA Core	<i>See also:</i>	
<i>Issues/Decisions:</i>			

EL4-VIL-SSS-REQ-01464		Maximum Failure Detection Time – SCI Connections to Object Controllers	(In negotiation)
The VIL shall detect any failure of an SCI connection to an Object Controller and perform an appropriate reaction within [TBD] seconds.			
<i>Comments:</i> The value depends on the interface timeout defined in the SCI interface specifications.			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL	<i>Verification:</i>	
	ECMUX	<i>See also:</i>	
<i>Issues/Decisions:</i>			

4.7 INTERNAL RASTA INTERFACES

EL4-VIL-SSS-REQ-01005		Internal RaSTA Interfaces - Station-Specific Safety Code	(Draft)
The VIL shall support station-specific configuration of the following RaSTA parameter for its internal RaSTA interfaces:			
Initial value of MD4 safety code			
It shall be ensured by suitable measures that different VIL instances never use the same initial value for the MD4 safety code.			
<i>Comments:</i> The RaSTA connections of both internal interface categories (ECMUX Command and Notification Interface, ECMUX Diagnosis Interface) shall use the same initial value.			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Technics (R&D)	<i>Origin:</i>	==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0015]
<i>Target Release:</i>	VIL_OBB_1.0.0	<i>Scope:</i>	Sub-System
<i>Allocation:</i>	VIL	<i>Verification:</i>	Analysis
	ECMUX		Integration test
	CAE Data Tools	<i>See also:</i>	
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0015 EC-MUX Command and Notification interface			
Decision: The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.			

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4.7.1 ECMUX COMMAND AND NOTIFICATION INTERFACE

EL4-VIL-SSS-REQ-00563 ECMUX Command and Notification Interface - RaSTA Connections (Draft)	
The internal Command and Notification Interface between components ELEKTRA Core and ECMUX shall be implemented by proprietary CHILL signal communication through two separate RaSTA connections (one for each diverse SW channel) per ECMUX instance.	
• RaSTA Version: 03.03 • Transport Layer: UDP (IPv4)	
<i>Comments:</i> For large stations the VIL will be configured to contain several ECMUX instances.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0015] <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0015 EC-MUX Command and Notification interface Decision: The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.	

EL4-VIL-SSS-REQ-01003 ECMUX Command and Notification Interface - Transport Channels (Draft)	
The RaSTA connections of the ECMUX Command and Notification Interface shall be available whenever at least one VIL redundant node is operational. Therefore, for each RaSTA connection and each redundant node of ECMUX, there shall be exactly one transport channel between that ECMUX node and the ELEKTRA Core node.	
<i>Comments:</i> On ELEKTRA Core side this interface is implemented by application CC.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0015] <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0015 EC-MUX Command and Notification interface Decision: The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.	

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4.7.2 ECMUX DIAGNOSIS INTERFACE

EL4-VIL-SSS-REQ-00564 ECMUX Diagnosis Interface - RaSTA Connections (Draft)	
The internal Diagnosis Interface between components ELEKTRA Core and ECMUX shall be implemented by proprietary CHILL signal communication through two separate RaSTA connections (one for each diverse SW channel of ECMUX). • RaSTA Version: 03.03 • Transport Layer: UDP (IPv4)	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0016] <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0016 EC-MUX Diagnosis Interface Decision: The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.	

EL4-VIL-SSS-REQ-01006 ECMUX Diagnosis Interface - Redundant Subchannels (Draft)	
The RaSTA connections of the ECMUX Diagnosis Interface shall be available whenever at least one VIL redundant node is operational. Therefore, for each RaSTA connection and each redundant node of ECMUX, there shall be exactly one transport channel between that ECMUX node and the ELEKTRA Core node.	
<i>Comments:</i> On ELEKTRA Core side this interface is implemented by application DGP.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> VIL ELEKTRA Core ECMUX	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 IDJ-RA] [EL4:IDJ-RA:ID:0016] <i>Scope:</i> Sub-System <i>Verification:</i> Analysis Integration test <i>See also:</i>
<i>Issues/Decisions:</i> EL4:IDJ-RA:ID:0016 EC-MUX Diagnosis Interface Decision: The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.	

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4.8 COMPONENT ELEKTRA CORE

4.8.1 GENERAL

EL4-VIL-SSS-REQ-00181		ELEKTRA Core - Fault Tolerance Architecture (Basic Redundancy)		(Draft)
In basic redundancy configuration, component ELEKTRA Core shall provide hot-hot redundancy using the TAS Platform NMR fault tolerance service in 2x2oo2 configuration. Thereby the ELEKTRA Core CG shall be robust against any HW failures affecting one CN.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation - Redundant Nodes ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00443] VIL Redundancy Concept - Availability	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	ELEKTRA Core	Verification:	Analysis Integration test	
		See also:		
Issues/Decisions:				

EL4-VIL-SSS-REQ-00182		ELEKTRA Core - Fault Tolerance Architecture (Georedundancy)		(Draft)
In georedundant configuration, component ELEKTRA Core shall provide hot-hot redundancy using the TAS Platform NMR fault tolerance service in 4x2oo2 configuration. Thereby the ELEKTRA Core CG shall be robust against any HW failures affecting one or two CNs.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00457] DIL Variants - Geo-Redundancy - Different Servers	
Target Release:	VIL_OBB_2.0.0	Scope:	Sub-System	
Allocation:	ELEKTRA Core	Verification:	Analysis	
		See also:	Integration test	
Issues/Decisions:				

EL4-VIL-SSS-REQ-01465		ELEKTRA Core – Reuse of CHILL Application SW		(Draft)
The ELEKTRA Core SW architecture shall allow reuse of existing CHILL application code by providing a CHILL runtime environment that supports processing in hot-hot redundancy mode.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	ELEKTRA Core	Verification:		
		See also:		
Issues/Decisions:				

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EL4-VIL-SSS-REQ-01001 ELEKTRA Core - Max. Number of Connected ECMUX (Draft)	
Component ELEKTRA Core shall be configurable to communicate with up to four instances of component ECMUX in order to control up to 1600 Object Controllers in total.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00610] Architecture - Number of OCs <i>Scope:</i> Sub-System <i>Verification:</i> Review Integration test Performance test <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-00229 ELEKTRA Core – Support Separate Application Restart (Draft)	
In order to reduce the impact of application SW failures, component ELEKTRA Core shall support the following restart behaviour in case of application failures:	
• A failure of SW application RCU shall lead to an automatic restart of RCU but not of CC and DGP. • A failure of SW application DGP shall lead to an automatic restart of DGP but not of CC and RCU. • A failure of SW application CC shall lead to an automatic restart of CC and RCU but not of DGP.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> Sub-System <i>Scope:</i> Integration test <i>Verification:</i> <i>See also:</i>
<i>Issues/Decisions:</i>	

4.8.2 PERSISTENT STATUS DATA (ELEKTRA CORE SPECIFIC)

EL4-VIL-SSS-REQ-01502 Persistent Status Data - Upload by both CEs of Master CN (Draft)	
Upon shutdown of the entire ELEKTRA Core computing group, only the CEs of the current master CN shall upload their locally stored persistent status files to a configurable remote object storage during shutdown.	
Each CE shall write and upload separate files. It shall be ensured that during reboot a CE0 cannot erroneously load a file written by CE1 and vice versa.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> Sub-System <i>Scope:</i> Integration test <i>Verification:</i> <i>See also:</i> ==> [EL4-VIL-SSS-REQ-01158] Persistent Status Data - Remote Storage
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-01159 Persistent Status Data - Early Check of Downloaded Files (Draft)	
After downloading persistent status data files from the remote object storage the VIL shall perform the following integrity checks: • safety code check (computed safety code must be equal to the one stored in safety code file) • cross-check between safety codes of the two CEs (must be identical) If any of these checks fails, all persistent status data and safety code files of the affected application shall be immediately deleted both locally and on the remote object storage.	
<i>Comments:</i> For the cross-check, only the safety code files of the other CE need to be downloaded and compared, not the actual persistent status data files.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-01158] Persistent Status Data - Remote Storage
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01161 Persistent Status Data - Check Consistency between Redundant Nodes (Draft)	
Upon application restart ELEKTRA Core shall check consistency of the reloaded persistent status data between the CNs. In case of inconsistency the data of the current master CN shall be considered as valid. Any slave CN with inconsistent persistent status data shall be restarted in order to trigger recovery of the data from the master CN.	
<i>Comments:</i> This check is implemented by TAS Platform NMR.	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-01158] Persistent Status Data - Remote Storage
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01162 Persistent Status Data - Inconsistent Values between Software Channels (Draft)	
For any application maintaining persistent status data and consisting of two diverse SW channels, both channels shall store and reload their persistent status data into separate files. Upon restart the application shall check consistency between the two channels for each individual status parameter. In case of inconsistency the VIL shall set the affected status parameters to the corresponding “default values after short downtime” as defined in EL4-DIL-SSS-REQ-00730 and generate a suitable diagnosis message.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> Sub-System <i>Verification:</i> Integration test <i>See also:</i> ==> [EL4-VIL-SSS-REQ-01500] Persistent Status Data - Inconsistent Values
<i>Issues/Decisions:</i>	

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EL4-VIL-SSS-REQ-01503 Database Generator - Include Identification and Version Data (Draft)	
The ELEKTRA Core database generator tool shall write the following data into the database key relation info file (keyrel.info): • Application name: text string (up to 40 characters) • Station ID: text string (up to 40 characters) • CMOS database schema version: <major>.<minor> • Station data version: <major>.<minor>	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> Sub-System <i>Scope:</i> Integration test <i>Verification:</i> ==> [EL4-VIL-SSS-REQ-01160] Persistent Status Data - Check Compatibility with Application Version ==> [EL4-VIL-SSS-REQ-01180] Persistent Status Data - Check Data Source ==> [EL4-VIL-SSS-REQ-01499] Persistent Status Data - Check Compatibility with Station Data <i>See also:</i>
<i>Issues/Decisions:</i>	

EL4-VIL-SSS-REQ-01504 Data Preparation - Increment CMOS Database Schema Version (Draft)	
The data preparation process shall ensure that the major number of the CMOS database schema version is incremented if at least one of the following conditions is true: • Order or number of CMOS relations has changed. • Order or number of elements of at least one CMOS relation has changed. • Number of records of at least one CMOS relation has changed. • Meaning of at least one element or element value has changed. • Default value of at least one element has changed. Otherwise only the minor version shall be incremented.	
<i>Comments:</i>	
<i>Category:</i> FUR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> VIL_OBB_1.0.0 <i>Allocation:</i> ELEKTRA Core CAE Data Tools	<i>Priority:</i> Mandatory <i>Origin:</i> Sub-System <i>Scope:</i> Integration test <i>Verification:</i> ==> [EL4-VIL-SSS-REQ-01160] Persistent Status Data - Check Compatibility with Application Version <i>See also:</i>
<i>Issues/Decisions:</i>	

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4.9 COMPONENT ECMUX

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EL4-VIL-SSS-REQ-00228		ECMUX Fault Tolerance Architecture (Georedundancy)		(In negotiation)
In georedundant configuration, component ECMUX shall provide hot-hot redundancy using the J4S fault tolerance service in 2x1oo1 configuration. Thereby component ECMUX shall be robust against any HW failures affecting one ECMUX redundant node. After such failure it shall be possible to start and integrate an ECMUX standby node in order to re-establish full connection redundancy.				
Comments: NOTE: This is a preliminary concept to provide that has not been finally decided.				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00457] DIL Variants - Geo-Redundancy - Different Servers	
Target Release:	VIL_OBB_2.0.0	Scope:	Sub-System	
Allocation:	ECMUX	Verification:	Analysis	
		See also:	Integration test	
Issues/Decisions:				

EL4-VIL-SSS-REQ-01002		ECMUX - Max. Number of Connected Object Controllers		(Draft)
Component ECMUX shall be able to control up to 400 Object Controllers.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Technics (R&D)	Origin:	==> [EL4 SSS] [EL4-DIL-SSS-REQ-00610] Architecture - Number of OCs	
Target Release:	VIL_OBB_1.0.0	Scope:	Sub-System	
Allocation:	ECMUX	Verification:	Review	
			Integration test	
			Performance test	
		See also:		
Issues/Decisions:				

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5. HISTORY

5.1 EDITION 01

First release of VIL SSS, prepared for the internal milestone VP7.
Corresponding DOORS baseline: EL4 SSS 1.3

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APPENDIX A DECISIONS

This chapter shall contain important decisions made in connection with the specified requirements.

EL4:IDJ-RA:ID:0013 VIL-ADM Diagnosis Interface	
Decision:	ADM is declared as responsible for diagnosis in the document Digitales Stellwerk - Technische Konzepte V04_PD01.
Status:	In negotiation
Decision Date:	02 September 2025
Priority:	Medium
Decision Authority:	Authors of Digitales Stellwerk - Technische Konzepte
Requirements:	EL4-SSS-REQ-00287, EL4-SSS-REQ-00289, EL4-SSS-REQ-00291, EL4-SSS-REQ-00292, EL4-SSS-REQ-00293, EL4-SSS-REQ-01307
Analysis:	ADM shall receive diagnostic messages from VIL (DGP) via Datalogger interface.

EL4:IDJ-RA:ID:0014 VIL VM Resource Management Interface	
Decision:	The VIL shall provide an interface for each of its virtual machines in order to monitor the VM's resource status.
Status:	In negotiation
Decision Date:	02 September 2025
Priority:	Medium
Decision Authority:	Architectural Expert team from VIL
Requirements:	EL4-SSS-REQ-01144
Analysis:	The data categories will be defined during the specification phase.

EL4:IDJ-RA:ID:0015 EC-MUX Command and Notification interface	
Decision:	The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.
Status:	In negotiation
Decision Date:	02 September 2025
Priority:	Medium
Decision Authority:	Architectural Expert team from VIL
Requirements:	EL4-SSS-REQ-00563, EL4-SSS-REQ-01003, EL4-SSS-REQ-01005
Analysis:	The VIL shall provide a command and notification interface between ELEKTRA Core and EC-MUX

EL4:IDJ-RA:ID:0016 EC-MUX Diagnosis Interface	
Decision:	The command and notification interface shall be based on two separate RaSTA connections and use internal CHILL signals.
Status:	In negotiation
Decision Date:	02 September 2025
Priority:	Medium
Decision Authority:	Architectural Expert team from VIL
Requirements:	EL4-SSS-REQ-00564, EL4-SSS-REQ-01006
Analysis:	The VIL shall provide a diagnosis interface between ELEKTRA Core and EC-MUX.

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